

COSMEON CLIMATE RISK ENGINE

Automated Intelligence Assessment

Post-event: 2022-06-01 to 2022-06-30 | Generated: 2026-03-01 06:55 UTC

Geographic Impact Area

Bounding Box Coordinates: [90.12, 25.31, 92.52, 26.55]

Executive Summary

Analysis of the study area (32,774 km2) utilizing multi-source satellite telemetry reveals 2465.0 km2 of flood extent and 14582.1 km2 of agricultural loss. Integration with demographic data indicates 1,486,427 individuals exposed. The probabilistic confidence peaked dynamically at 97.8%. The derived Severity Index is 33.6/100.

Key Quantitative Metrics

Flooded Area	2464.98 km2	SAR VV threshold -2.0 dB
Vegetation Damage	14582.13 km2	NDVI drop threshold -0.12
Exposed Population	1,486,427 ppl	WorldPop 100m Integration
Anomalous Flood	2355.31 km2	JRC Predictive History mapping
Peak Confidence	97.8%	Ensemble fusion overlap (>95%)
Severity Index	33.6/100	Dynamic 5-factor composite

Data Provenance

Sentinel-1 SAR Base	22 scenes (2022-05-01 - 2022-05-31)
Sentinel-1 SAR Post	18 scenes (2022-06-01 - 2022-06-30)
Sentinel-2 Opt Base	44 scenes (Cloud limit: 90%)
Sentinel-2 Opt Post	21 scenes (Cloud limit: 90%)
Processing Scale	150 m spatial resolution